📚 Basic Yocto Questions (Knowledge & Understanding)

1. What is the Yocto Project and what are its core components?
2. What is the difference between OpenEmbedded and Yocto?
3. Explain the role of bitbake in the Yocto build system.
4. What are layers in Yocto, and why are they important?
5. What is a recipe (.bb file) in Yocto? Explain its key fields.
6. How does Yocto handle dependencies between recipes?
7. What is the difference between PACKAGECONFIG, DEPENDS, and RDEPENDS?
8. What are DISTRO_FEATURES and how do they affect the build?
9. How can you add a custom layer in Yocto?
10. What are machine configurations in Yocto? Give examples.

Intermediate Questions (Customization & Troubleshooting)

11. How do you add a new package to an existing Yocto image?
12. What is the difference between IMAGE_INSTALL and CORE_IMAGE_EXTRA_INSTALL?
13. Explain the difference between do_configure, do_compile, and do_install.
14. How can you override a task (e.g., do_compile) in a Yocto recipe?
15. How do you debug a recipe that fails to fetch a source tarball?
16. How does Yocto handle source mirrors and local file storage?
17. How can you make BitBake fetch source from a specific Git branch or commit?
18. What is devtool and how can it help during development?
19. How do you apply a patch to a recipe in Yocto?
20. Explain inherit in Yocto and give examples (e.g., autotools, cmake, systemd).

⚙️ Advanced Questions (Integration, BSP, and Real-World Use)

21. How would you create a custom image with only specific packages?
22. How do you build and integrate a custom kernel in Yocto?
23. Explain how you would add support for a new hardware board in Yocto.
24. How do you generate an SDK for application development using Yocto?
25. What is EXTRA_OECONF and how is it used?
26. How does Yocto manage root file system generation and compression?
27. Explain the purpose of tmp and sstate-cache directories.
28. How do you reduce the image size in Yocto?
29. Describe the steps to include a new init system (e.g., systemd) into your image.
30. What are the ways to implement Over-The-Air (OTA) updates in Yocto-based systems?

🔍 Debugging & Performance Tuning

31. How do you use bitbake -e and what insights can it give?
32. How do you diagnose issues in a slow Yocto build?
33. What causes a BitBake task to re-run when you didn't change anything?
34. How do you cache and reuse build artifacts across teams/machines?
35. Explain how BB_NO_NETWORK and BB_HASHBASE_WHITELIST help debugging.

🗀 Real-World Scenarios / Systems Questions

36. If a package is not appearing in your final rootfs image, how would you troubleshoot it?
37. Your image boots but network does not come up — how do you debug?
38. Explain the differences between using a prebuilt SDK vs building from Yocto tree.
39. How do you handle license compliance in a Yocto-built system?
40. How do you generate and deploy signed images in Yocto?

📃 File System, Init, Bootloader, and Deployment

41. How do you create a dual-partition setup for A/B OTA with Yocto?
42. How do you include and configure U-Boot for a custom board?
43. What's the role of .wic files and how do you customize them?
44. How would you integrate TPM or Secure Boot into a Yocto-based system?
45. How do you set default system services using systemd in Yocto?

▯ Mixed Stack and Open Questions

46. How do you integrate third-party binary blobs (e.g., GPU firmware) into a Yocto image?
47. How do you test a Yocto-built image using QEMU?
48. How do you cross-compile a C++ application outside the Yocto tree but link it to the rootfs?
49. What would be your approach to create a container-based Yocto system?
50. How do you maintain multiple products and hardware variants using the same Yocto tree?

- ✅ **Google** – Focus on deep technical understanding, debugging, architecture, tool internals, and optimization.
- ✅ **Amazon** – Focus on scalable system design, modularity, security, OTA updates, CI/CD, and infrastructure ownership.

▯ Google-Style Yocto Interview Questions

Deep system internals, debugging, reproducibility, open-source practices

🔧 Tooling & Internals

1. What happens internally when you run bitbake core-image-minimal?
2. How does BitBake determine whether a task needs to be rebuilt?

3. Explain how sstate cache works. What are its advantages and failure modes?
4. What is BB_HASHBASE_WHITELIST used for?
5. How does Yocto implement reproducible builds?
6. How do bitbake -e and bitbake -g help in understanding build internals?

🔍 Debugging & Root Cause Analysis

7. A package builds correctly but is missing in the rootfs. How do you debug this?
8. What tools do you use to trace do_fetch or do_compile failures?
9. How do you find the origin of a variable like SRC_URI when it is overridden?
10. Why might a previously successful build fail in CI/CD with no recipe changes?

🧠 Kernel & Bootloader Customization

11. How do you override the kernel version and add a custom patchset in Yocto?
12. What is the structure of a Yocto-compatible U-Boot recipe? How do you add a boot script?
13. How can you debug kernel boot logs from a QEMU image?

🗁 File System & Configuration

14. How do IMAGE_FEATURES, DISTRO_FEATURES, and PACKAGECONFIG interact?
15. Explain the layer priority and recipe override mechanism (BBLAYERS, PREFERRED_PROVIDER, etc.).
16. How does Yocto manage init systems, and what happens if two conflicting services are enabled?

📜 Advanced Concepts

17. How do you reduce build time in a Yocto CI setup across multiple branches?
18. How do you test and simulate a new BSP using QEMU before porting to hardware?
19. How do you handle host contamination and ensure isolated builds?
20. In what scenarios would you use a multiconfig setup?

🏢 Amazon-Style Yocto Interview Questions

Scalability, ownership, OTA, infrastructure, multi-product management

📦 Product Ownership & Architecture

1. How do you manage multiple product variants (board types) from the same Yocto source tree?
2. What is your strategy to modularize feature sets across devices with different capabilities?
3. How do you structure custom layers for reusable product features?

🚀 OTA & Deployment

4. Describe your approach to A/B system updates using Yocto.
5. How would you build and deploy secure OTA updates with image signing?
6. What Yocto tools or methods can you use to verify firmware integrity on the target?

📈 CI/CD & Scalable Build Systems

7. How do you integrate Yocto builds into a CI/CD pipeline (e.g., Jenkins, GitHub Actions)?
8. How do you cache Yocto build artifacts across multiple build servers?
9. How do you detect regressions in rootfs size, boot time, or package count during CI?

🔐 Security & Compliance

10. How do you track licenses of all included packages in a Yocto build?
11. What is the role of LICENSE, LIC_FILES_CHKSUM, and COPY_LIC_DIRS in recipes?
12. How would you implement Secure Boot in a Yocto project?

🌐 Connectivity and Cloud-Integration

13. How would you bundle cloud agent software into a Yocto image with configuration?
14. What strategy would you use to ensure unique identity and security credentials per device?

🔲 Real-World Customization

15. How do you create a Yocto image with minimal boot time for a production environment?
16. How do you make a Yocto image work on multiple memory configurations (e.g., 256MB, 512MB)?
17. What is the difference between using core-image-full-cmdline and a custom image?
18. How do you integrate external closed-source binaries into your Yocto image?